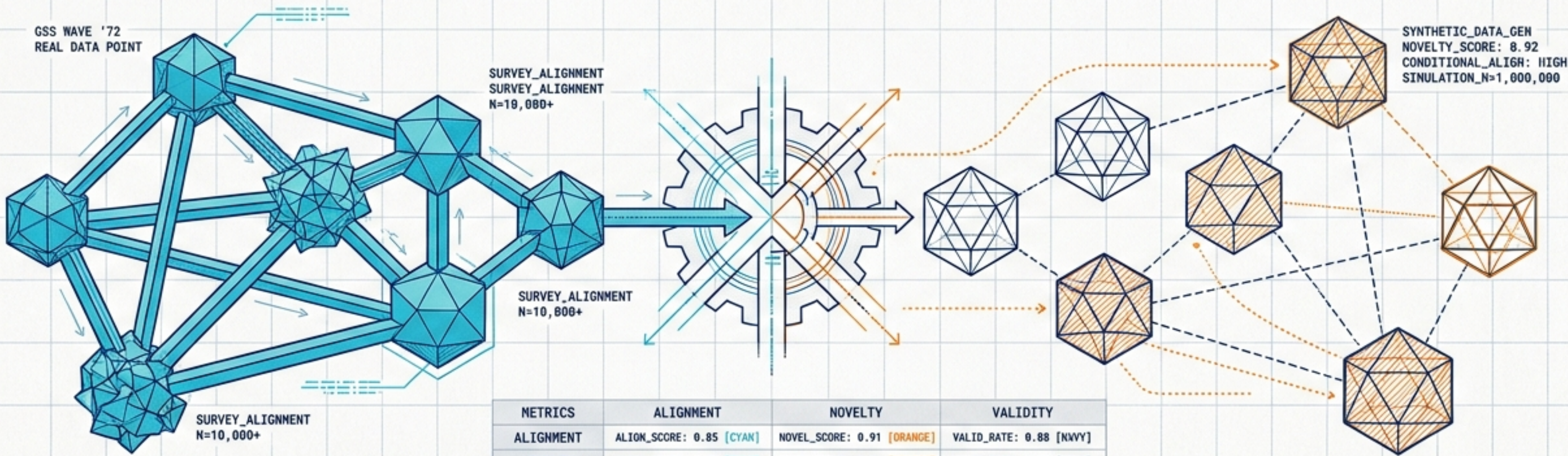


Technical deep-dive and empirical benchmark
across 34 GSS Waves (1972–2024).

Evaluating Synthetic Social Data

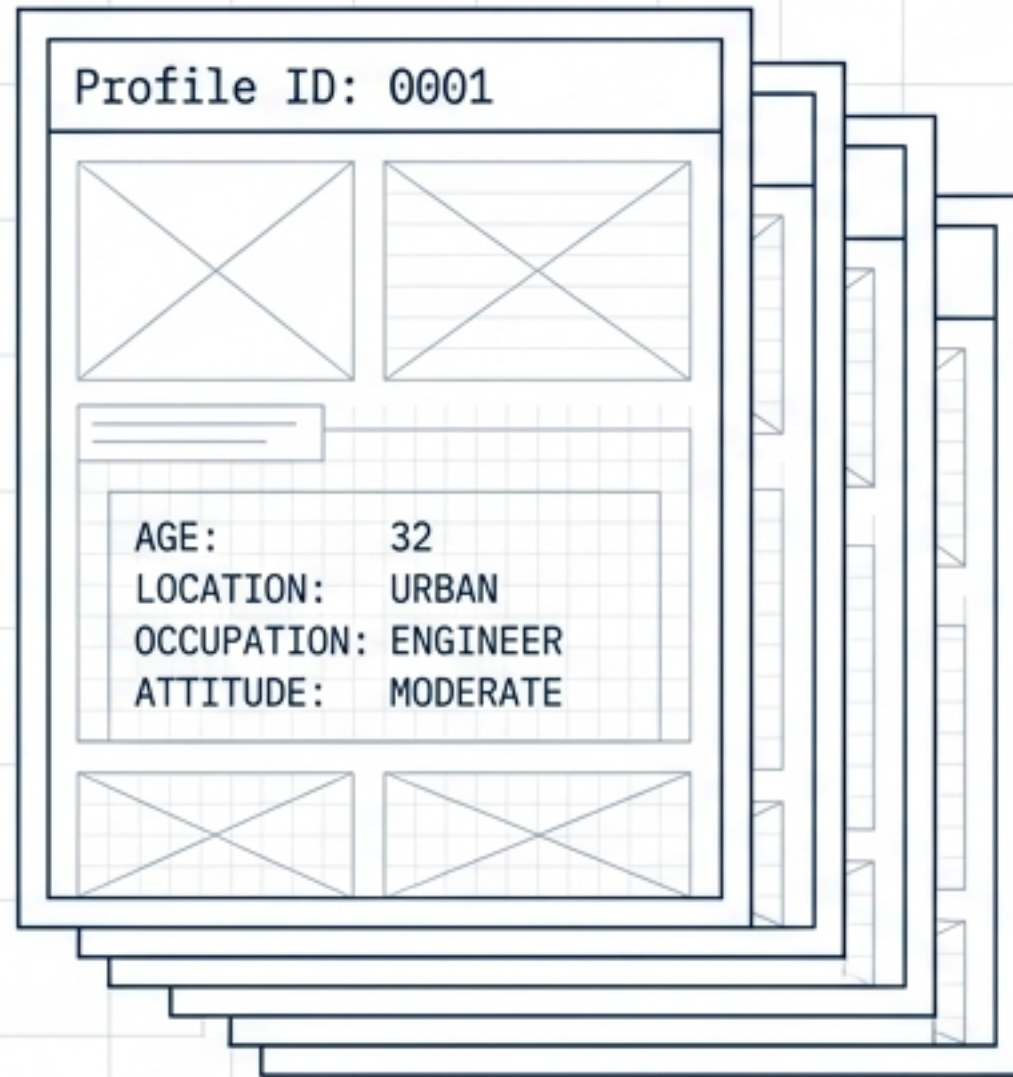
A Framework for Conditional Alignment and Novelty



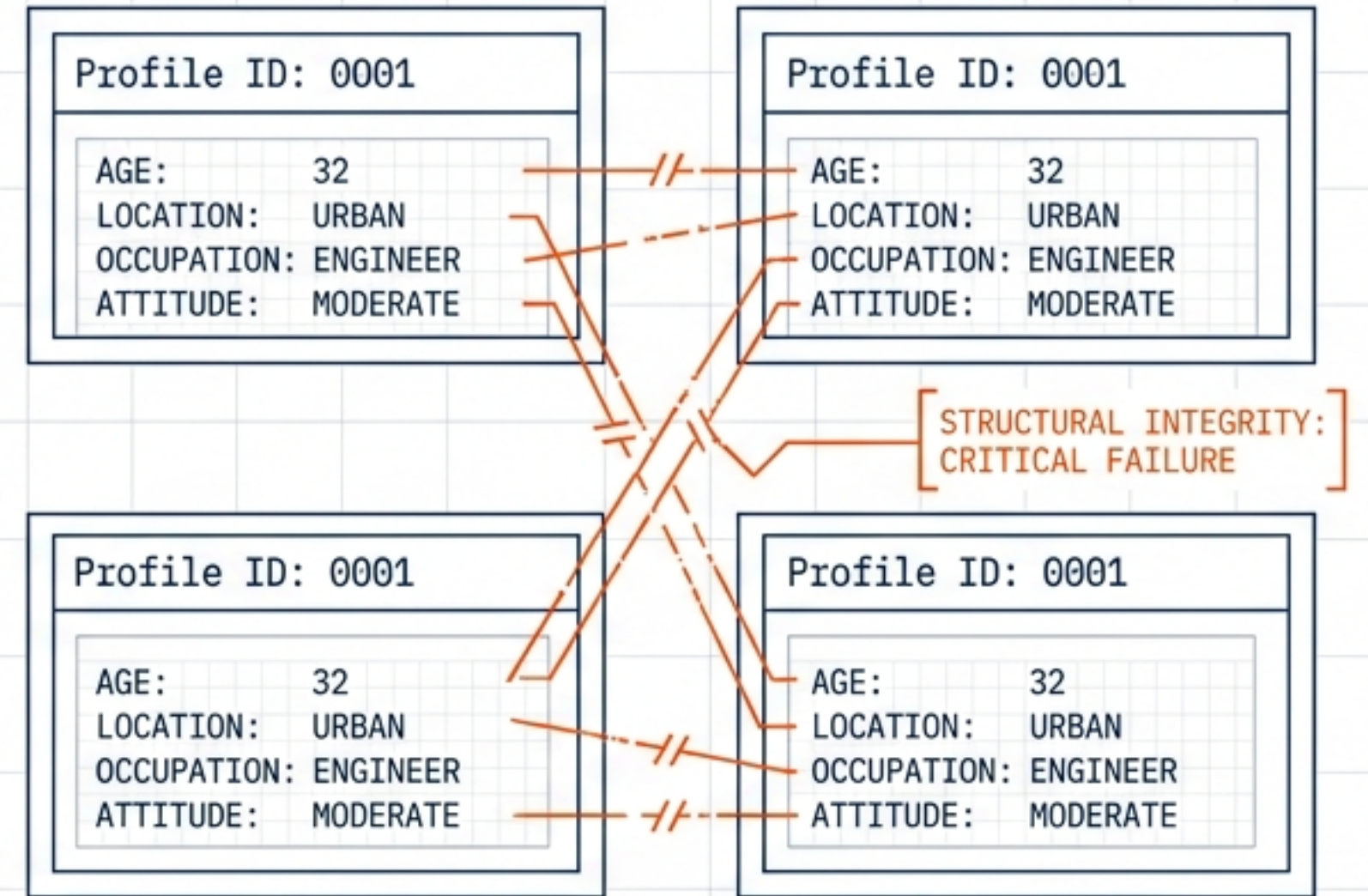
The Generative Dilemma in Social Data

Plausible Individual Records

✓
[REALISM: HIGH]



Failing to Reproduce Empirical Dependencies



How do we test if a synthetic population captures structural social attitudes without just rewarding models that memorize and copy observed respondents?

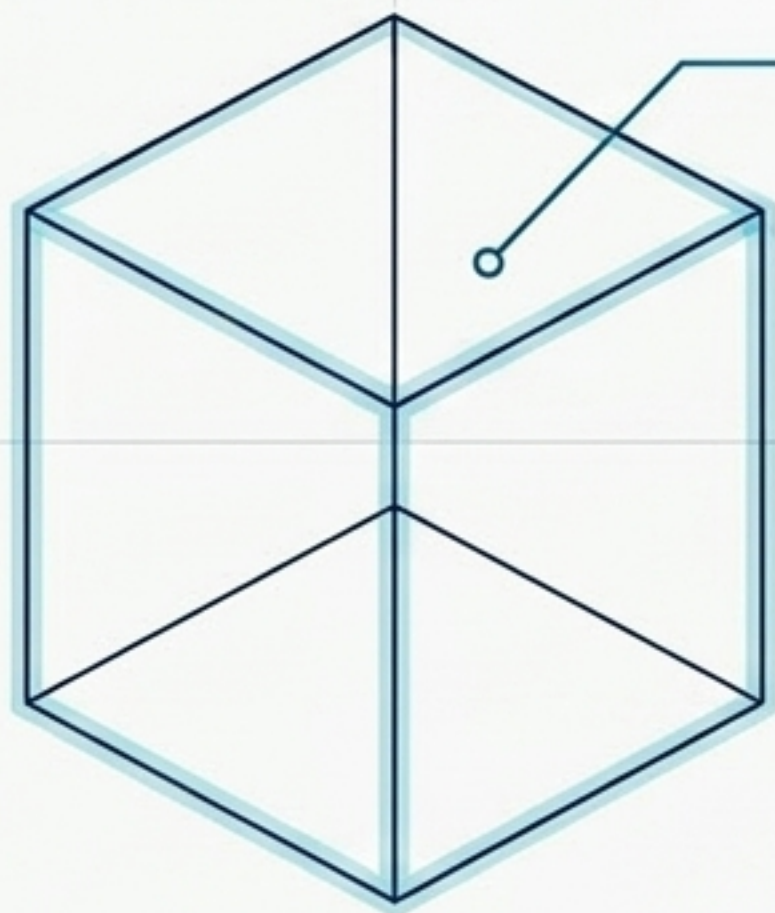
The Validation Blindspot

Approach	Method	The Blindspot
Likelihood-based	ELBO, perplexity	Requires the generator to have a tractable joint density.
Task-based	Train on synthetic, test on real	Depends on subjective downstream tasks; obscures structural mismatches.
Embedding-based	FID, two-sample tests	Relies on feature extractors unrelated to the specific data domain.

Comparisons of component-wise moments or covariance matrices are also **insufficient**. They capture pairwise linear structure but miss nonlinear dependencies.

Why Marginals and Covariance Lie

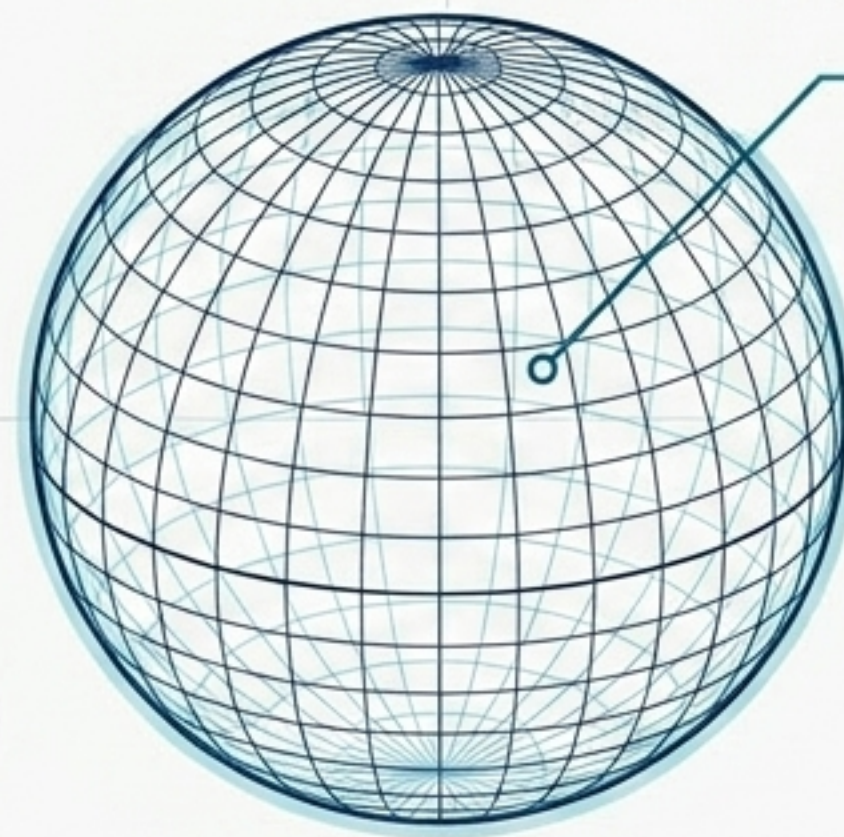
Uniform Distribution



Flat
conditional
density

=

Gaussian Distribution



Normal
conditional
density

Identical first and
second-order moments

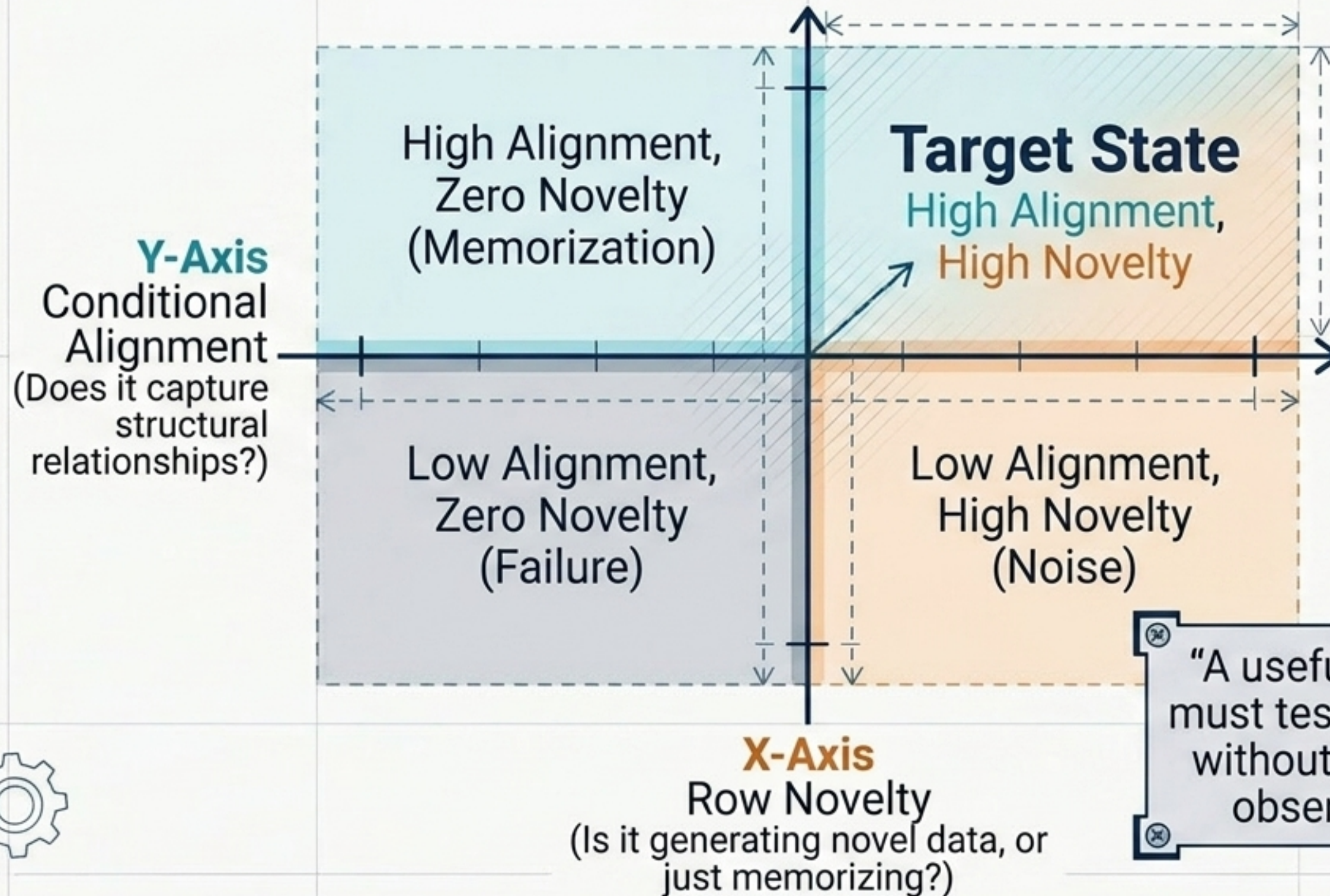
$$\mu = 0, \text{Cov} = I_3$$

Uniform MAP-alignment = 1

Gaussian MAP-alignment = 0.707

Identical moments and covariance matrices can mask fundamentally different conditional structures.

The Two-Axis Validation Framework



Dissecting the MAP-Alignment Score

Learned Conditional Kernel:
The observed probability evaluated under the real-data conditional.

$$v(x, i) = \frac{\phi_i(x_i | x_{-i})}{\max_y \phi_i(y | x_{-i})}$$

Maximum Probability:
The modal probability in that exact same respondent context.

Score = 1 only when the value is the model's conditional mode.

Theoretical Guarantee: **Identifiability**

The Brook-Dobrushin Factorization



Reference Configuration



Iterative Conditioning

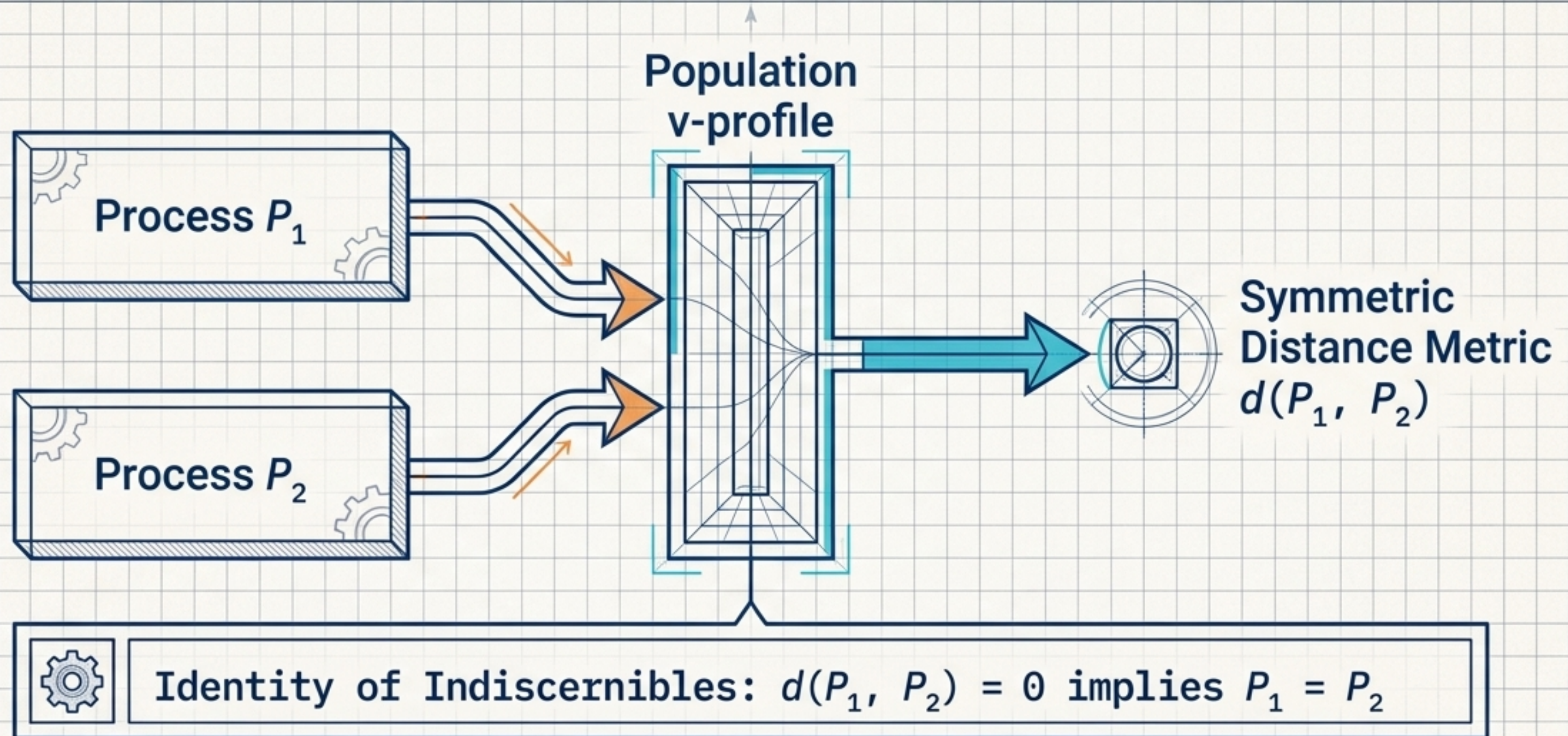


Fully Reconstructed
Joint Distribution

$$\frac{P(x)}{P(x_0)} = \prod_i \frac{P_i(x_i | x_{<i}, x_{>i,0})}{P_i(x_{i,0} | x_{<i}, x_{>i,0})}$$

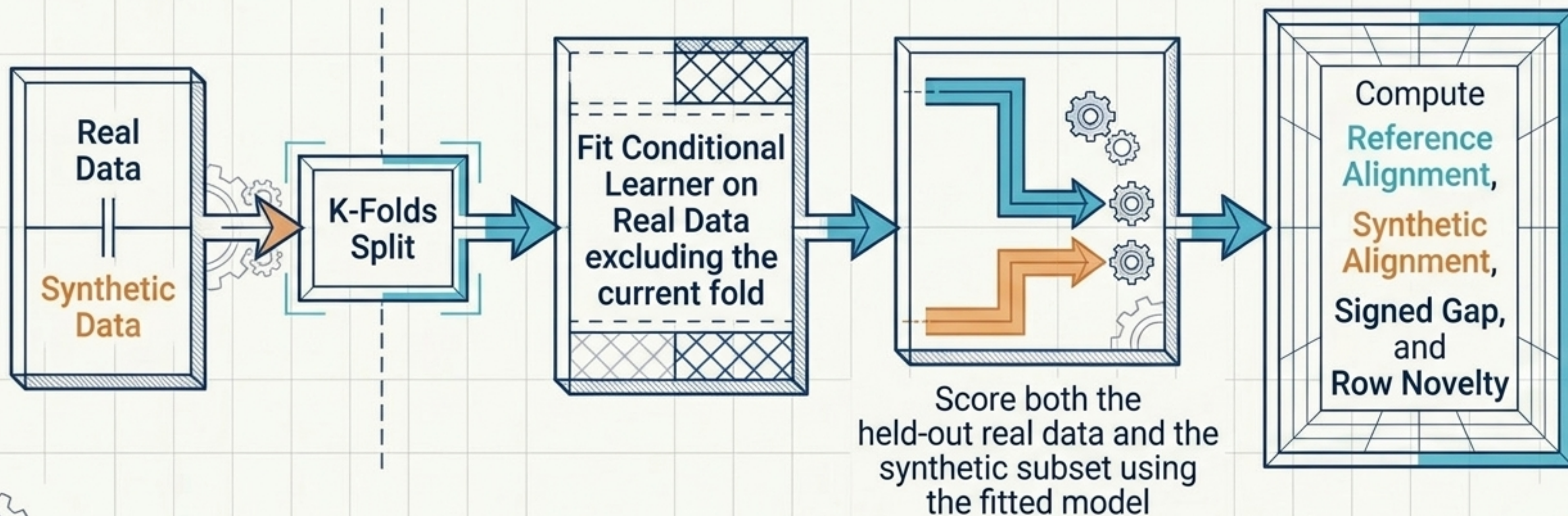
Under strict positivity and compatibility, a complete system of full conditionals perfectly reconstructs and identifies the joint distribution ($P = Q$).

Distance Between Generative Processes



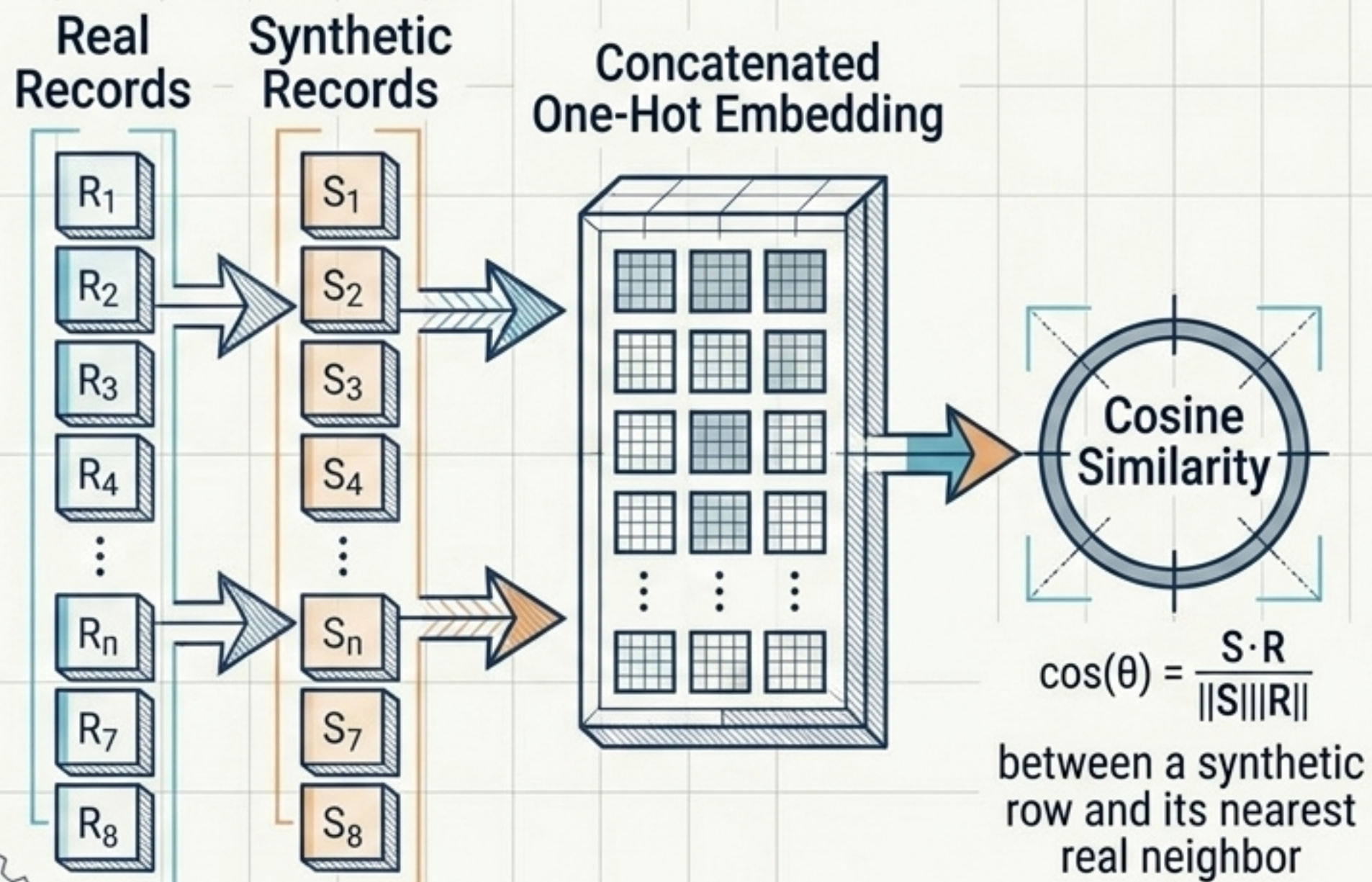
The complete normalized conditional profile induces a **mathematically sound, symmetric metric** on strictly positive finite-state generative processes.

Algorithm 1: Cross-Fitted Evaluation



Holding out evaluator data, training conditional inference trees, and scoring on an even playing field without data leakage.

Defining Row Novelty



Row Novelty measures separation from observed respondents, ensuring high structural alignment isn't achieved via simple resampling.

The GSS Empirical Benchmark Setup

34

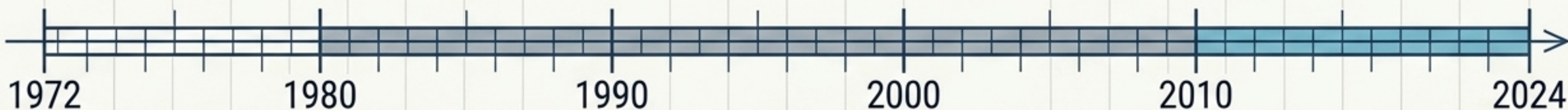
Survey Waves

71,667

Respondent Records

197 to 1,273

Modeled Variables per wave



GSS Data
Collection

50% LSM & Evaluator Training

50% Held-out Analysis Split

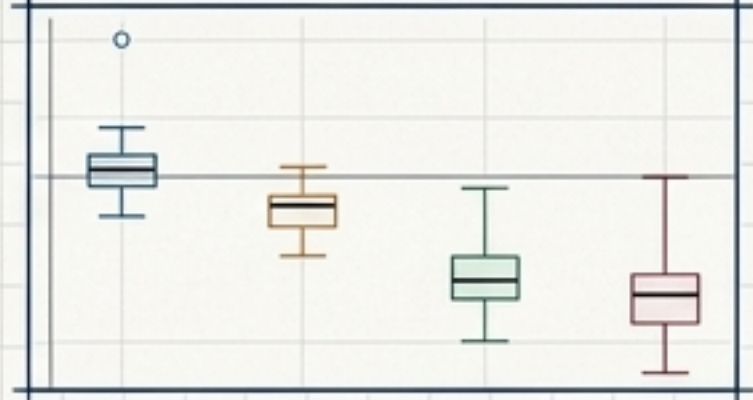


Validating against the General Social Survey
(1972–2024) cross-sections.

Benchmark Performance Matrix

Generator	Mean MAP Alignment	Signed Gap	Excess Alignment	Row Novelty
LSM	0.8032	+0.0027	1.062	0.6200
Original (Control)	0.8004	0.0000	1.000	0.0000
Chow-Liu	0.7872	-0.0132	0.698	0.3440
CTGAN	0.7630	-0.0375	0.143	0.4335
Ind. Baseline	0.7567	-0.0437	0.000	0.4381

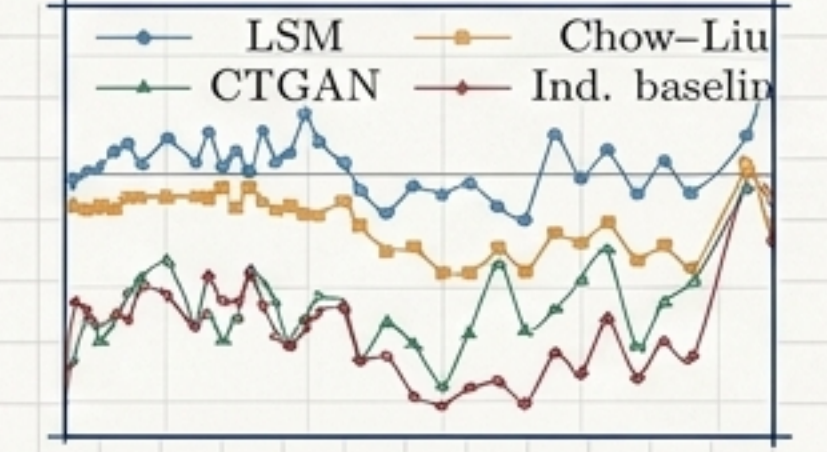
Signed MAP-Alignment Gap



Row Novelty Across Waves

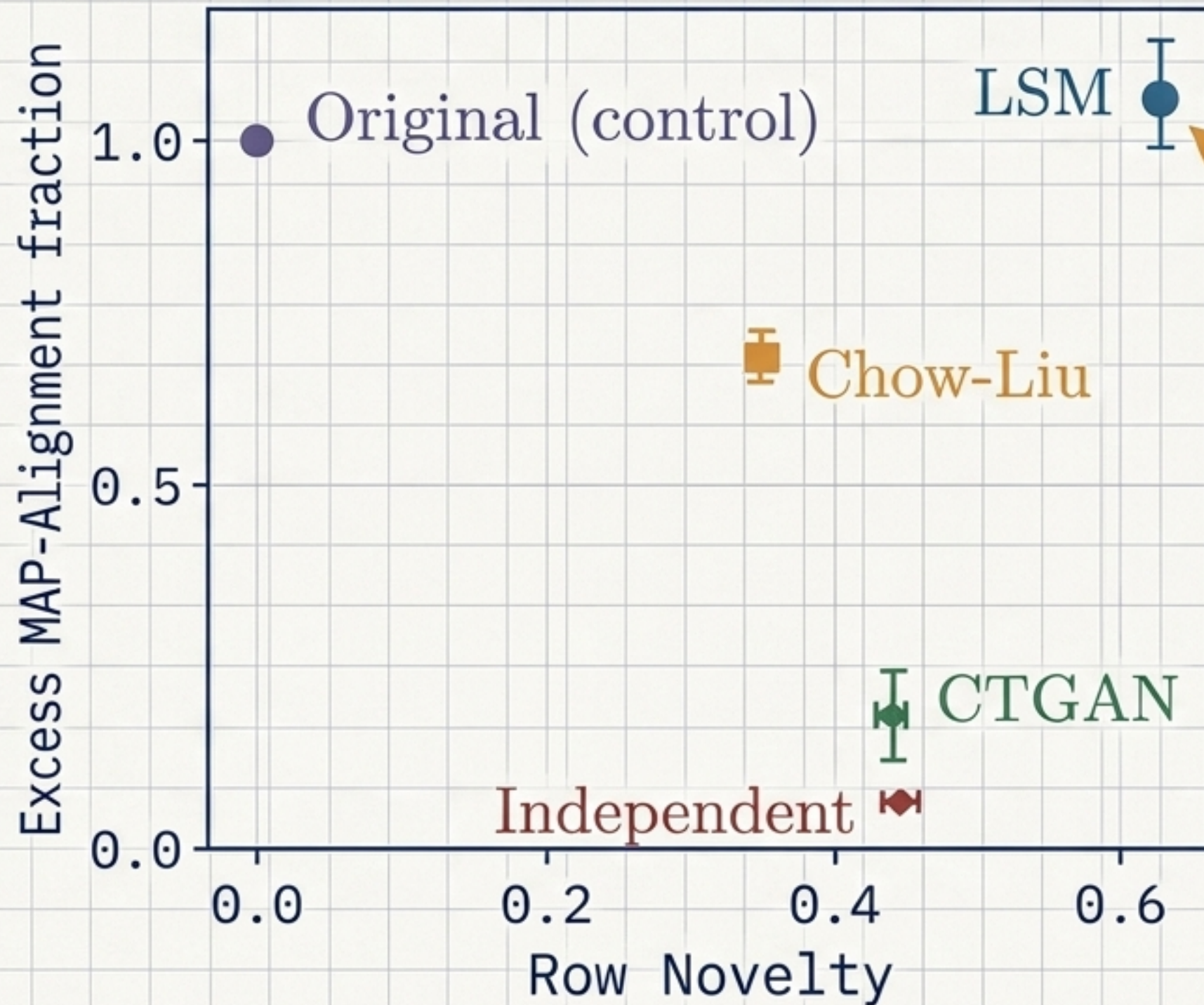


Wave-Wise MAP-Alignment Retention



LSM retains control-level MAP alignment while providing high row novelty across 34 waves.

The Alignment-Novelty Frontier



The desired regime:
The desired regime:
combining reference-level
MAP alignment with
separation from observed
respondents.

Practical Implementation for Data Scientists

Checking Structural Fidelity

Checking Memorization

Compute MAP-Alignment

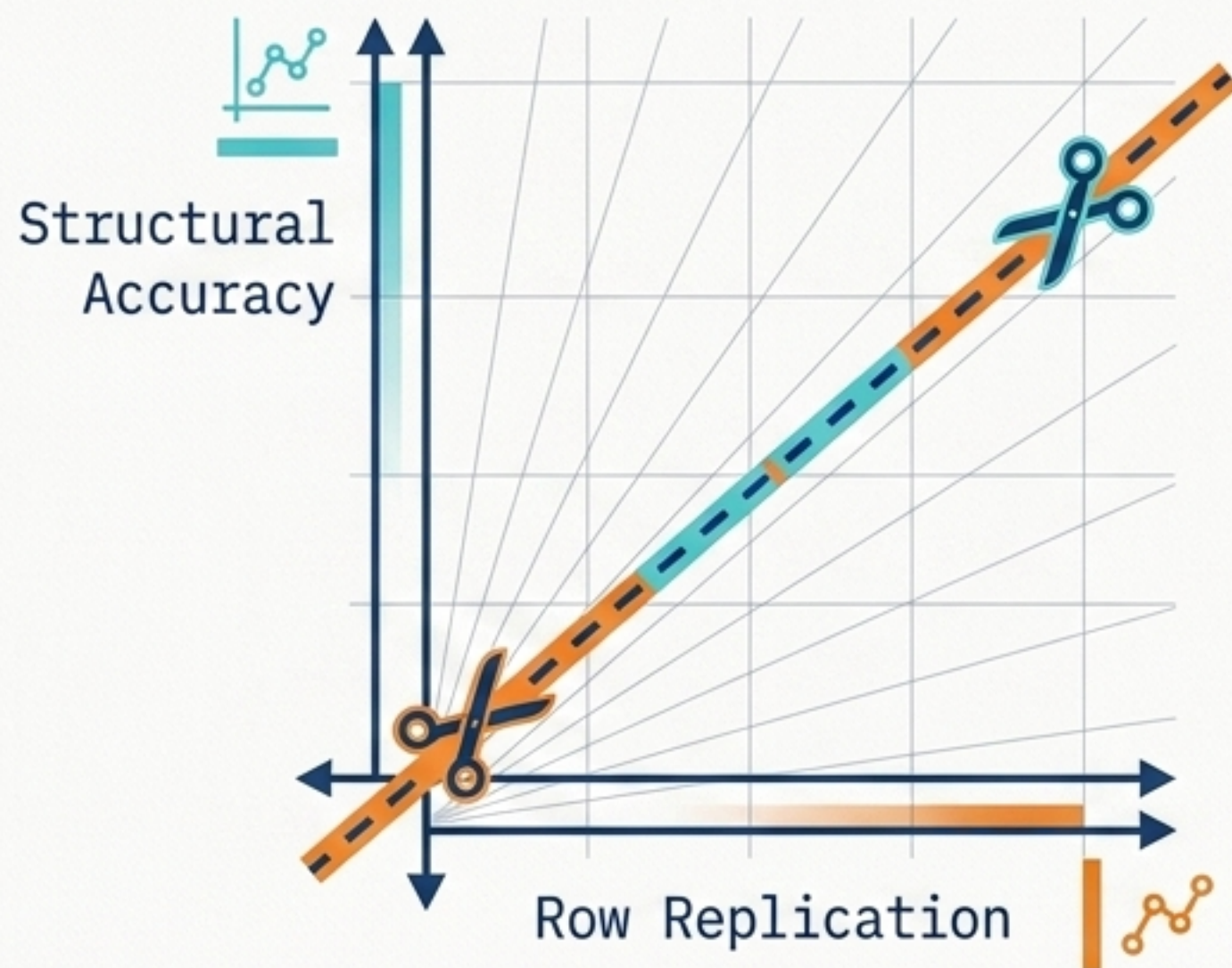
Compute Nearest-Real Similarity

Report **signed** MAP-alignment
gap & zero-anchored **excess
alignment fraction**.

Report upper-tail summaries
(**95th/99th percentile**) &
exact duplicate rates.

Note: Do not replace application-specific validation
or formal privacy/disclosure-risk analysis.

The Ultimate Conclusion



Synthetic data validation fundamentally requires decoupling structural accuracy from row replication. The MAP-alignment framework provides a model-agnostic layer to guarantee both.

Terminal

```
> pip install lsynth  
> github.com/zeroknowledgediscovery/lsynth
```

Access the full implementation via the `lsynth` Python package for computational social science.